Development of a Framework to Detect Fake Offers on E-commerce Websites Using Rule-Based and Machine Learning Techniques

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Abstract

The proposed dissertation describes how a hybrid system can be created to identify fake offers on e-commerce sites through a combination of rule-based heuristics and supervised machine learning in MATLAB. The paper discusses the increasing problem of fraudulent promotional practices such as false discounts, price comparison exaggeration and false sense of urgency that contribute to the lack of trust in online markets by consumers. An end-to-end literature review reveals the shortcomings of single-purpose rule systems, including inflexibility and non-adaptability, and machine learning models, which frequently demand massive labelled datasets as well as have problems with interpretability. In order to eliminate these gaps, a two-layered detection framework is suggested. The first stage involves domain-driven rules to filter suspicious offers according to logical thresholds, including abnormal discount percentages, and mismatched address, and unusual time of transaction. At the second step, the use of supervised learning models, specifically the Random Forest classifier, is performed to train on a dataset provided by Kaggle (more than 300,000 labeled records). Data preprocessing, feature engineering, categorical encoding and class imbalance were utilized in its implementation. Quantitative analysis in terms of accuracy, precision, recall, F1-score, confusion matrices, and ROC curves have shown that the hybrid methodology is better than the standalone models. The accuracy of the Random Forest was 95.21% and the composite framework enhanced false positive and negative detection as well as robustness since it identified the unrecognizable fraudulent patterns. The research is valuable in that it presents the practicality of the MATLAB toolboxes to detect fraud and highlights the importance of hybrid models as a way of balancing flexibility and interpretability. Weak points are dataset imbalance, and changing fraud strategies, and further studies are suggested in the context of real-time adaptation, deep learning, and automatic rule refinement.

Acknowledgement

I would like to express my gratitude to my supervisor whose guidance and support in this research process was priceless. I would also like to thank my department to support me academically and Kaggle to provide the dataset utilized in the present work. Last but not least, I owe much to my family and peers to be constantly motivated and supportive of my work.

List of Acronyms

Acronym	Full Form
AI	Artificial Intelligence
AUC	Area Under the Curve
CNN	Convolutional Neural Network
F1-Score	Harmonic Mean of Precision and Recall
ML	Machine Learning
RF	Random Forest
ROC	Receiver Operating Characteristic
SHAP	SHapley Additive exPlanations
SVM	Support Vector Machine

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Chapter 1: Introduction

In this research project, the researcher aims at developing a two-level framework that identifies bogus offers in an e-commerce venue. We will work with MATLAB as the major data processing, rule creation as well as development of machine learning models. Models will be trained by using a publicly available data source based on Kaggle.

1.1 Aim and Objective

This project is dedicated to developing and applying the intelligent finding framework with the help of combining rule-based logic and supervised machine learning to find fake promo on digital stores.

The objectives of the project are:

- To understand trends of false and real offers on a marked dataset
- To develop heuristics based on the norms of identify abnormal discount patterns
- In MATLAB training of classification models to improve accuracy of prediction

1.2 Problem Statement

The problem of false marketing is popular on E-commerce sites through falsified deals, such as wrong discounts, distorted comparisons, and deceitful pricing. Such fraudulent advertisements do not only victimize consumers but also decrease the confidence in the web-based trading (Salman and Balaram, 2025). Current detection systems tend to be reactive and do not have an ability to discover the dynamic patterns in different groups.

1.3 Research Questions

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		June					July			August			Sept
Task	Duration	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10	Week 11	Week 12
Supervisor Allocation	7 Days												
Library session	7 Days												
Milestone 1:- Topic selection	3 Days												
Proposal preparation and review	11 days												
Milestone 2:- Proposal Submission	1 day												
Abstarct	2 Days												
Chapter 1- Introduction	7 Days												
Chapter 2- Literature Review	8 Days												
Milestone 3:- Practical implementation/ Model Development	20 days												
Chapter 3- Methodology	4 days												
Chapter 4- Experimental Results Documentation	7 days												
Chapter 5- Conclusion and Future work	4 Days												
Final review and Modifications	3 days												
Milestone 4:- Final Submission	1 day												

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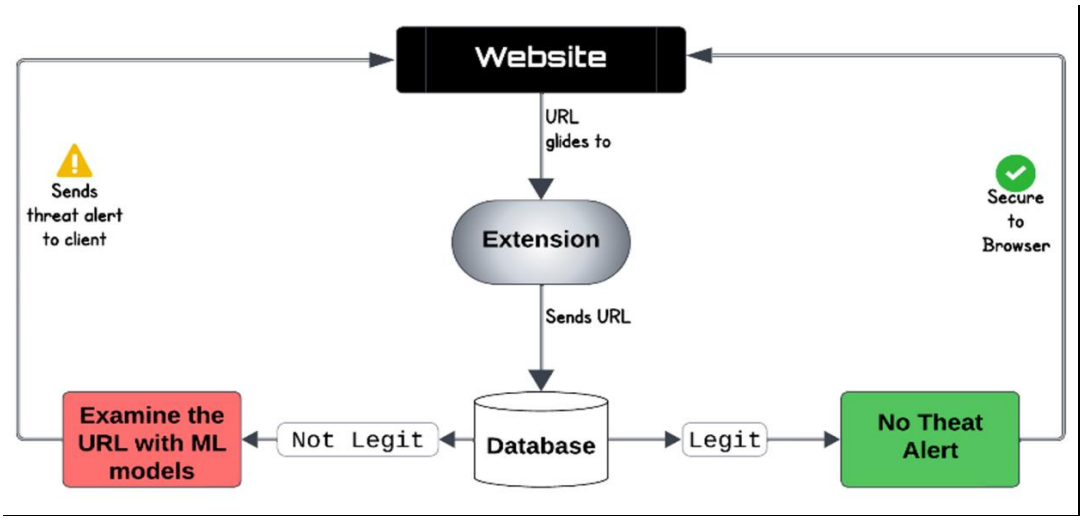

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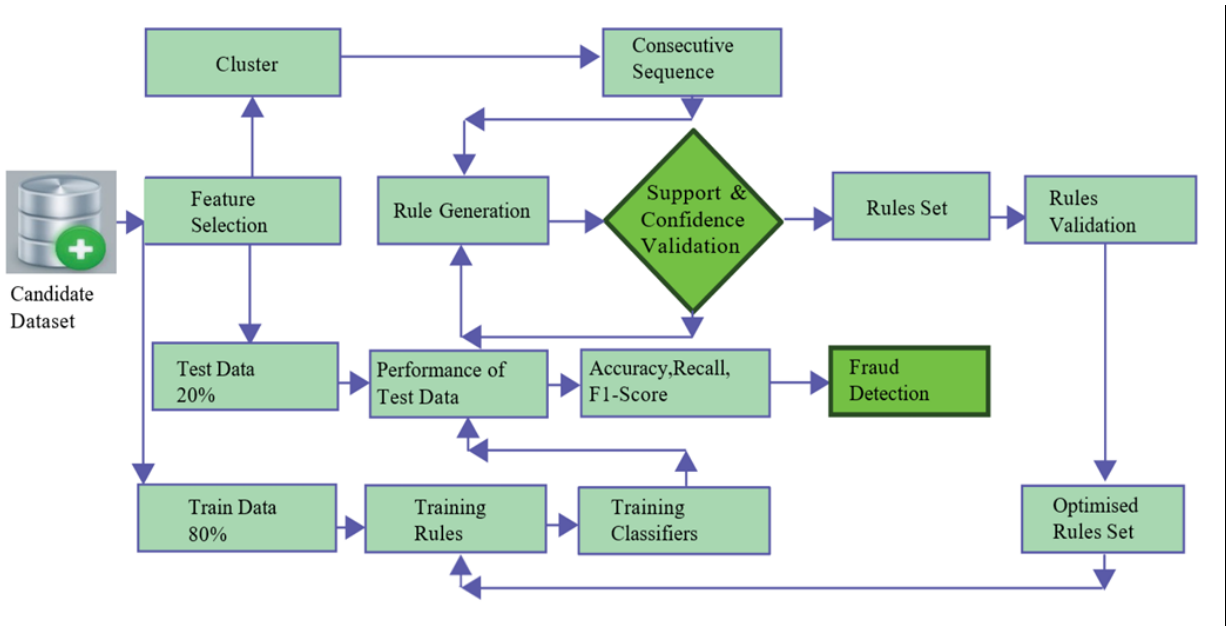

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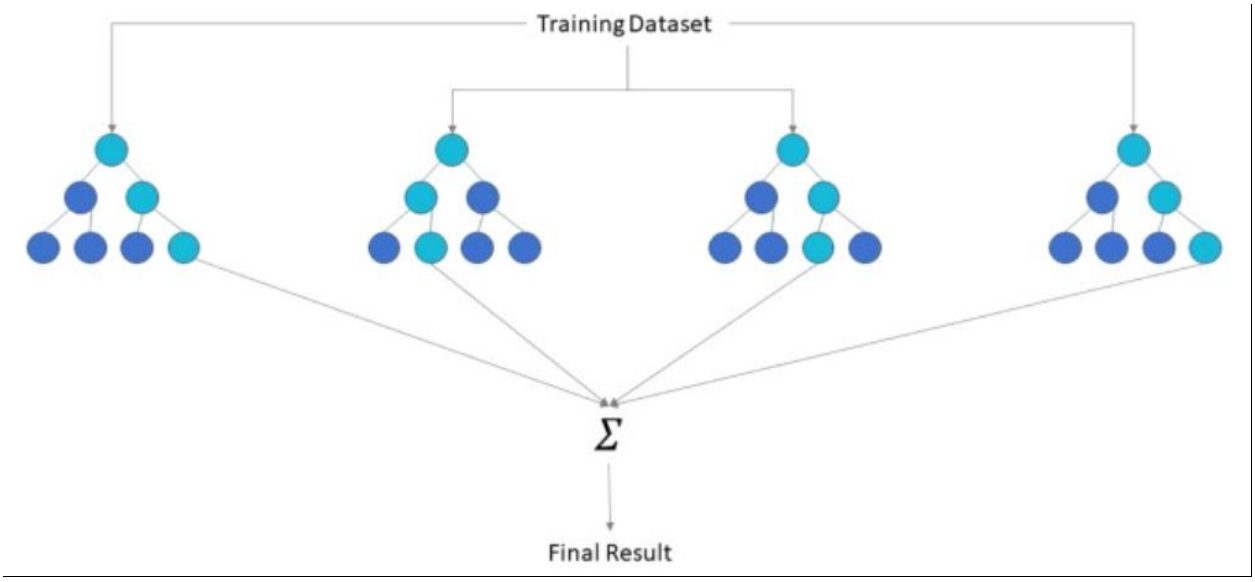
Training Dataset
Σ
Final Result

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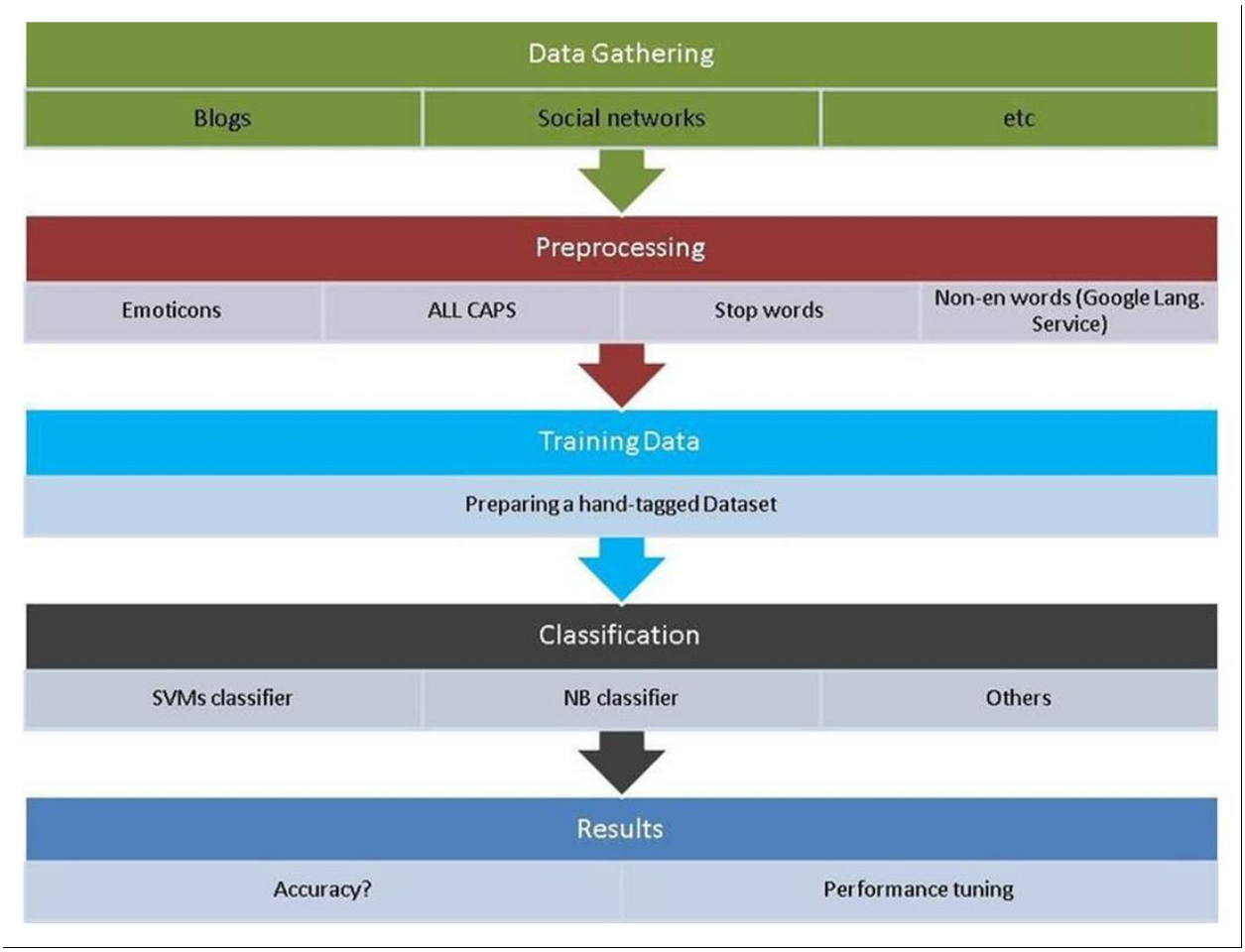

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Author(s) & Year	Technique	Specification	Limitation	Dataset	Results
Mutemi and Bacao (2024)	Systematic Review (ML Techniques)	Comprehensive review of ML methods for e-commerce fraud detection	No experimental model; lacks validation	70+ academic papers reviewed	Identified 6 core ML methods; mapped trends & gaps in e-commerce fraud detection
Islam et al. (2025)	ML (Multiple Models)	Prediction using synthetic e-commerce datasets and ML classifiers	Synthetic data reduces real-world generalizability	Synthetic e-commerce transaction dataset	Achieved 87% accuracy on synthetic data using XGBoost and SVM models
Chy and Buadi (2024)	ML + Browser Extension	Real-time fraud detection with web plugin and backend ML scoring	Prototype stage; lacks comparative evaluation	Website usage and fraud score logs (implied)	Proposed real-time scoring plugin; no metrics reported; qualitative system prototype
Devarmani et al. (2025)	Qualitative + Descriptive Study	Analysis of fake discount tactics and consumer awareness	No technical model or automation implementation	User survey + observational sources	Found 68% of users misled by fake discounts; proposed awareness tools
Yang, Hu and Zhu (2025)	Rule-Based + Ensemble (Belief Rule Base)	Fraud detection using belief rule-based ensemble models	High computational complexity; rule tuning needed	Online payment transaction records	Improved fraud detection by 12% over baseline models using hybrid BRB model
Islam et al., (2024)	Rule-Based + ML (Hybrid)	Combines business rules with ML to detect financial fraud	Requires quality rule design for accuracy	Financial fraud transactional data	F1-Score: 0.93; Accuracy: 91%; model interpretable through rules
Reddy et al. (2024)	ML + Big Data Analytics	Big data pipelines for e-commerce fraud analytics	Depends on massive datasets; high processing cost	Large-scale e-commerce dataset	AUC: 0.94; Over 17% gain in detection efficiency with real-time analytics
Peddinti et al. (2024)	Deep Learning (Anomaly Detection)	Deep learning model for transaction-level anomaly detection	Low explainability; requires GPUs and long training	E-commerce and finance transaction logs	Recall: 90%; Anomalies flagged with minimal false positives
Jeyachandran (2024)	Comparative Review	Comparison of fraud prevention techniques in platforms	No implementation; focused on literature synthesis	Multiple secondary case studies	Summarized strengths and weaknesses of 15 fraud detection techniques
Chatrath et al. (2022)	ML (CNN for Images)	Product image-based deception detection	Focus limited to visual deception only	E-commerce product image datasets	Model Accuracy: 88%; Detected deceptive imagery with CNN classifier

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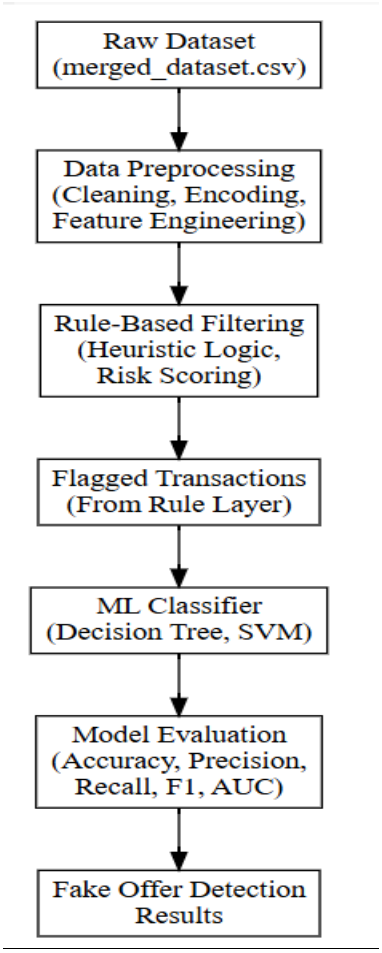
Raw Dataset
(merged_dataset.csv)
Data Preprocessing
(Cleaning, Encoding,
Feature Engineering)
Rule-Based Filtering
(Heuristic Logic,
Risk Scoring)
Flagged Transactions
(From Rule Layer)
ML Classifier
(Decision Tree, SVM)
Model Evaluation
(Accuracy, Precision,
Recall, F1, AUC)
Fake Offer Detection
Results

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```
% Display the shape of the dataset
fprintf('Dataset Loaded: %d rows and %d columns\n', height(data), width(data));

% Display first 5 rows
disp('First 5 rows of the dataset:');
disp(data(1:5,:));

% Show summary
disp('Dataset Summary:');
summary(data);
```

```
%% Step 2: Data Cleaning & Preprocessing
fprintf('\n--- Data Cleaning & Preprocessing Started ---\n');
```

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```
% Remove missing Transaction_Date
rowsBefore = height(data);
data = rmmissing(data, 'DataVariables', {'Transaction_Date'});
fprintf('Removed %d rows with missing Transaction_Date\n', rowsBefore - height(data));

% Remove unrealistic customer ages
rowsBefore = height(data);
data = data(data.Customer_Age >= 10 & data.Customer_Age <= 100, :);
fprintf('Removed %d rows with invalid Customer_Age\n', rowsBefore - height(data));
```

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```
% Convert categorical columns
categoricalCols = {'source','browser','sex','Payment_Method','Product_Category','Device_Used'};
for i = 1:length(categoricalCols)
    data.(categoricalCols{i}) = categorical(data.(categoricalCols{i}));
end
fprintf('Converted %d columns to categorical\n', length(categoricalCols));

% Derived features from Transaction_Date
data.Transaction_Weekday = weekday(data.Transaction_Date);
data.Is_Weekend = ismember(data.Transaction_Weekday, [1 7]);
fprintf('Derived features: Transaction_Weekday & Is_Weekend\n');

% Standardize numerical features
numericalCols = {'Transaction_Amount','Customer_Age','Account_Age_Days','Transaction_Hour','Quantity'};
for i = 1:length(numericalCols)
    col = numericalCols{i};
    data.([col '_z']) = (data.(col) - mean(data.(col))) / std(data.(col));
end
fprintf('Standardized %d numerical columns\n', length(numericalCols));
```

```
%% Step 3: Rule-Based Filtering
fprintf('\n--- Rule-Based Filtering ---\n');

rule1 = data.Transaction_Amount > 500 & data.Account_Age_Days < 30;
rule2 = data.Transaction_Hour >= 0 & data.Transaction_Hour <= 5;
rule3 = data.Address_Match == 0;
rule4 = abs(data.Transaction_Amount_z) > 2;

data.Rule_Score = double(rule1) + double(rule2) + double(rule3) + double(rule4);
data.Rule_Based_Flag = data.Rule_Score >= 2;

totalFlagged = sum(data.Rule_Based_Flag);
fprintf('Total suspicious transactions flagged by rules: %d out of %d\n', totalFlagged, height(data));

% Display sample flagged rows
disp('Sample flagged transactions:');
disp(data(data.Rule_Based_Flag == 1, 1:10));
```

```
%% Step 4: ML Classifier Training
fprintf('\n--- Machine Learning Classifier Training ---\n');

features = [data.Transaction_Amount_z, data.Customer_Age_z, data.Account_Age_Days_z, ...
            data.Transaction_Hour_z, data.Quantity_z, data.Address_Match];
X = double(features);
Y = data.Is_Fraudulent;

% Split data
cv = cvpartition(Y, 'HoldOut', 0.30);
XTrain = X(training(cv), :);
YTrain = Y(training(cv), :);
XTest  = X(test(cv), :);
YTest  = Y(test(cv), :);
fprintf('Training Set: %d | Test Set: %d\n', size(XTrain,1), size(XTest,1));

% Train Random Forest
rng('default');
model_RF = fitcensemble(XTrain, YTrain, 'Method', 'Bag', 'NumLearningCycles', 50, 'Learners', 'Tree');
fprintf('Random Forest Model Trained with %d Trees\n', 50);
```

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Sample flagged fraud predictions:

Transaction_Date	Transaction_Amount	Customer_Age	Is_Fraudulent	Account_Age_Days	Transaction_Hour	source	browser	sex	Payment_Method
2024-03-28 20:34:40	2206.2	39	1	7	20	Ads	IE	M	PayPal
2024-04-02 21:10:41	1302.2	33	1	10	1	Ads	FireFox	F	debit card
2024-02-26 15:58:50	805.75	42	1	16	1	SEO	IE	F	bank transfer
2024-02-21 07:56:34	1153.4	43	1	5	3	SEO	FireFox	F	bank transfer
2024-01-01 12:11:05	3437	38	1	9	4	Ads	IE	M	debit card
2024-03-09 12:14:59	1491.5	31	1	4	0	SEO	Safari	F	PayPal
2024-02-04 22:47:57	745.75	40	1	25	2	Direct	FireFox	M	bank transfer
2024-02-29 04:26:12	897.88	30	0	29	4	Ads	Opera	M	credit card
2024-03-13 18:57:40	651.75	24	1	1	2	Direct	Chrome	M	PayPal

```
--- Applying ML to Rule-Flagged Transactions ---
Transactions flagged for ML analysis: 1526
Predicted fraudulent transactions among flagged: 725 out of 1526
Sample predicted fraudulent transactions:
```

Transaction_Date	Transaction_Amount	Customer_Age	Is_Fraudulent	Account_Age_Days	Transaction_Hour	source	browser	sex	Payment_Method
2024-03-28 20:34:40	2206.2	39	1	7	20	Ads	IE	M	PayPal
2024-03-11 06:51:10	809.01	46	0	16	6	Direct	Chrome	M	credit card
2024-04-02 21:10:41	1302.2	33	1	10	1	Ads	FireFox	F	debit card
2024-02-26 15:58:50	805.75	42	1	16	1	SEO	IE	F	bank transfer
2024-02-21 07:56:34	1153.4	43	1	5	3	SEO	FireFox	F	bank transfer
2024-01-01 12:11:05	3437	38	1	9	4	Ads	IE	M	debit card
2024-03-09 12:14:59	1491.5	31	1	4	0	SEO	Safari	F	PayPal
2024-02-04 22:47:57	745.75	40	1	25	2	Direct	FireFox	M	bank transfer
2024-01-04 20:03:44	816.95	32	1	10	5	Ads	IE	M	bank transfer
2024-02-15 15:13:15	2918.2	14	1	17	0	Direct	Safari	M	credit card
2024-01-01 03:46:54	598.6	18	1	12	3	SEO	Chrome	F	credit card
2024-01-06 03:53:31	795.55		6	Ads	FireFox	M	debit card		
2024-03-22 08:25:28	810.16	26	1	21	8	SEO	Safari	M	credit card

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```
--- Machine Learning Classifier Training ---
Training Set: 209340 | Test Set: 89716
```

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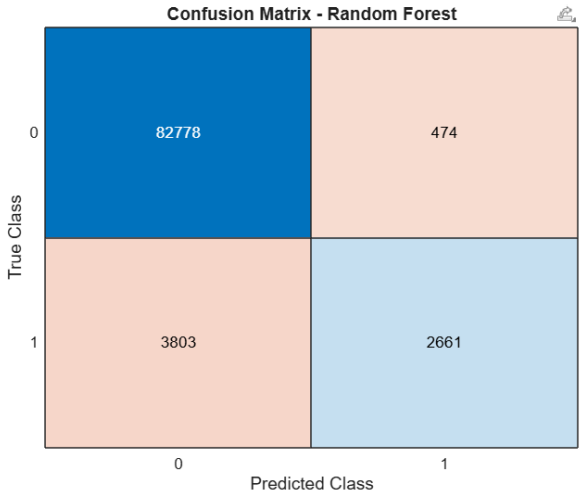
```
--- Machine Learning Classifier Training ---
Training Set: 209340 | Test Set: 89716
Random Forest Model Trained with 50 Trees
Confusion Matrix:
       82758         494
        3853        2611

Accuracy: 95.15%
Precision: 84.09%
Recall: 40.39%
F1 Score: 54.57%
```



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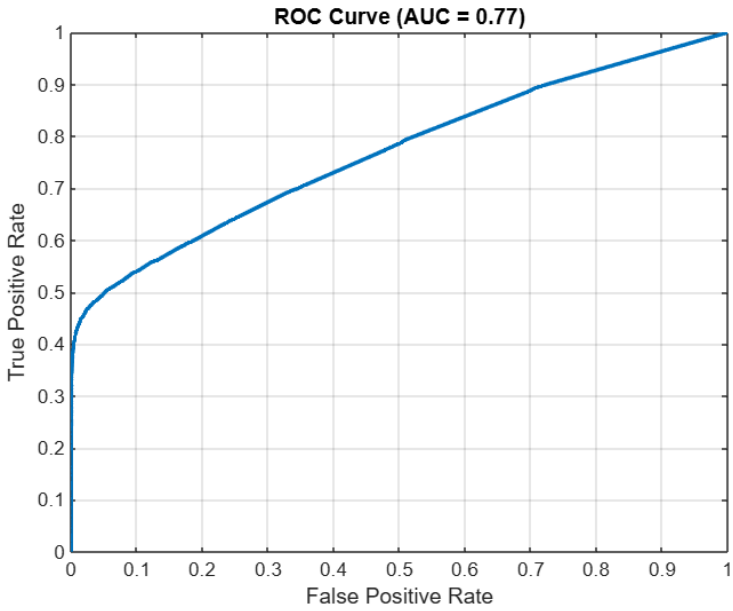

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```
Trained model saved as Shanmukhi_RF_Model.mat
>> Press Ctrl + Shift + P to generate code with Copilot
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